



TavaOne Education

General License Course · 2023–2027 Pool

Module 6

Capstone & Exam Prep

Full Pool Review · 2 Practice Exams · You're Ready



Exam Day Logistics

WHERE

Right here — VEs come to us at the Pinellas STEM Radio Lab

BRING

Government-issued photo ID · Your existing Technician license

FRN

Bring your FCC Registration Number if you have one; VEs can look you up if not

FEE

\$35 FCC fee — paid ONLINE through CORES after the VEs submit your application. NOT paid at the exam.

FORMAT

35 questions · multiple choice · 4 options · 26 correct to pass (74%)



What's on the Exam

Subelement	Topic	Exam Questions
G1	Commission's Rules	5
G2	Operating Procedures	5
G3	Radio Wave Propagation	3
G4	Amateur Radio Practices	5
G5	Electrical Principles	3
G6	Circuit Components	2
G7	Practical Circuits	3
G8	Signals and Emissions	3



Practice Exam 1

35 Questions · 30 Minutes · Silent Individual Work



Review

Go Through Every Missed Question



Hour 2

Targeted Review + Practice Exam 2



Quick Reference: Most-Tested Facts

Rules & Operating (G1/G2)

60m: channelized, USB only, 100W ERP max
30m: no phone, 200W max
CEPT/IARP: General and above only (not Tech)
VE minimum: 3 VEs per session
General VEs: Element 2 only
PRB-1: zoning must reasonably accommodate
LSB below 10 MHz, USB above 10 MHz
QRL = is this freq in use?
ARES = ARRL volunteer; RACES = govt activated

Technical (G5/G7/G8/G9)

$X_c = 1/(2\pi fC)$ — decreases with frequency
 $X_L = 2\pi fL$ — increases with frequency
Full-wave rectifier ripple = 120 Hz
Class A/AB = linear (SSB); Class C = FM/CW only
J3E = SSB suppressed carrier
FM deviation: ± 5 kHz max; capture effect
Dipole = $468/f(\text{MHz})$ feet
SWR 2:1 = 11% reflected power
ATU does NOT fix feed line SWR
SSB duty cycle $\approx 20\%$; FM = 100%



Quick Reference: Propagation & Safety (G3/G0)

Propagation

D layer: absorbs daytime, gone at night

E layer: sporadic-E, unpredictable

F2 layer: the DX layer, reflects HF

MUF: above it = goes to space

LUF: below it = absorbed by D layer

High SFI = good HF; high K-index = bad

NVIS: 80m/40m nearly vertical, regional

Gray line: enhanced DX at dawn/dusk

Safety

Filter caps hold charge after power off

Bleeder resistor discharges caps safely

Lightning: DISCONNECT antennas

Gas discharge arrestors: close but not enough

RF MPE: SSB $\approx$ 20% duty cycle

FM = 100% duty cycle for RF exposure

Controlled vs. uncontrolled MPE limits differ

Distance is most effective exposure reducer



You're Ready.

You did the work. You know this material.

After you pass:

- VEs submit your paperwork → pay \$35 at CORES ([fcc.gov](https://www.fcc.gov)) → license appears in ULS in a few days
- Once Active in ULS → you can operate immediately, no paper license needed
- What opens up: 80m, 40m, 20m phone · DX chasing · POTA on HF · HF contesting